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CAPTAIN: a multimodal foundation model pretrained on co-assayed single-cell RNA and protein

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Proteins act as the terminal effectors of cellular function, encoding the phenotypic consequences of genomic and transcriptomic programmes. Although transcriptomic profiles serve as accessible proxies, they remain incomplete surrogates for the proteomic landscape that ultimately defines cellular phenotypes. Current single-cell foundation models, however, are trained exclusively on transcriptomes, resulting in biased and partial characterizations of cellular states. To address this limitation, we introduce CAPTAIN, a multimodal foundational model pretrained on over four million single cells with concurrently measured transcriptomes and a curated repertoire of 382 surface proteins across diverse human and mouse tissues. Our results show that CAPTAIN learns unified multimodal representations by modelling cross-modality dependencies and capturing the diversity of cellular states across complex biological contexts. CAPTAIN generalizes robustly across both fine-tuning and zero-shot settings, excelling in core downstream tasks such as protein imputation and expansion, cell type annotation, and batch harmonization. Beyond improved accuracy in multi-omics integration, CAPTAIN generates novel hypotheses regarding protein-driven intercellular dynamics, including potential immune interaction patterns linked to COVID-19 severity.

Single-cell sequencing has transformed our ability to define cell states and resolve cellular heterogeneity across biological systems[1–4]. Through large-scale initiatives such as the Human Cell Atlas, hundreds of millions of individual cells have been profiled, yielding expansive transcriptomic landscapes spanning tissues, developmental trajectories, disease conditions, and experimental perturbations[5–8]. This exponential growth of high-dimensional data has catalyzed the development of foundation models[9,10], which are large-scale self-supervised architectures with self-attention mechanisms trained on massive single-cell corpora[11]. Emerging models such as scGPT[12], Geneformer[13], GeneCompass[14], CellPLM[15] and scFoundation[16], alongside novel architectures exploring federated learning for privacy-preserving applications[17], leverage the sequential structure of gene expression data, akin to language models in natural language processing, to distil semantic representations of cell identity that generalize across downstream single-cell tasks[18–22]. While these models have

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demonstrated strong performance in tasks such as cell type classification, gene expression imputation, and perturbation response modelling, their reliance on transcriptomic data alone often does not fully capture protein-level variation, particularly in immunological and developmental contexts where surface proteins serve as key phenotypic determinants[23,24]. This unimodal paradigm, although scalable and widely accessible, imposes inherent limitations on our ability to resolve cellular immunophenotypes and capture the full complexity of cell states.

Current proteomic profiling methods, with Cellular Indexing of Transcriptomes and Epitopes by Sequencing (CITE-seq) being the most widely adopted, enable the simultaneous measurement of transcriptomes and surface proteins[25,26]. However, their utility is constrained by the limited and heterogeneous composition of cell surface protein (CSP) panels, which vary substantially between studies and typically encompass only tens to a few hundred targets[27]. Compounded by challenges in cross-modality integration, batch effects from technical and biological sources, and dataset harmonization, these limitations hinder the interpretability of cellular states and the resolution of subtle phenotypic differences[28–31]. These challenges underscore the urgent need for a multimodal modelling framework that integrates transcriptomic and proteomic signals at single-cell resolution, thereby enabling biologically grounded representations that generalize across systems, species, and analytical tasks.

Here, we present CAPTAIN, a multimodal single-cell foundation model jointly pretrained on harmonized transcriptomic and surface proteomic profiles. CAPTAIN is designed to bridge gene expression and protein-level phenotypes by learning transferable joint RNA-protein representations at single-cell resolution. By integrating transcriptomic and proteomic signals during large-scale pretraining, CAPTAIN enables protein-aware cell representations that generalize across datasets, modalities and downstream analytical tasks.

CAPTAIN adopts a representation-centric formulation for multimodal single-cell modelling, in which surface proteins are treated as first-class components of a shared latent space rather than as auxiliary prediction targets. Through protein-aware pretraining on a large-scale corpus of co-assayed RNA-protein profiles (scT & P-4M), transcriptomic states and surface protein phenotypes are aligned into unified cell embeddings. Leveraging cross-modal attention, CAPTAIN integrates transcriptomic and proteomic features to produce protein-informed representations that support zero-shot and fine-tuned protein inference within a fixed surface-protein vocabulary, facilitate integration across datasets with heterogeneous antibody panels, and enable downstream analyses that depend on proteomic context, including protein-informed cell-cell communication inference and immune marker discovery. Our main contributions are as follows:

- Scale and design of the multimodal training data: CAPTAIN is pretrained on the largest curated multimodal single-cell RNA-protein co-assayed dataset to date (scT & P-4M; 4.2 million cells spanning 13 tissues). The dataset is systematically organized through standardized preprocessing, harmonized protein naming, and functional categorization of the surface protein vocabulary. This shared and curated surface protein panel of 382 targets enables consistent modelling of surface proteins across studies, including proteins that are not directly linked to a single gene, such as multi-subunit surface receptors.
- Protein imputation guided by a shared surface protein panel: CAPTAIN enables protein abundance imputation by leveraging large-scale multimodal pretraining on paired RNA-protein data. By learning systematic correspondences between transcriptomic states and surface protein expression within a shared surface protein panel, the model supports robust inference of protein levels (significantly outperforming 5 state-of-the-art methods in prediction accuracy, $p < 0.05$) in cells where proteins are not directly measured, without relying on dataset-specific reference mappings.
- Protein-aware downstream analyses enabled by the joint RNA-protein representations: Building on its protein imputation capability, CAPTAIN supports downstream analyses that depend on proteomic context even in the absence of direct protein measurements. These include fine-grained cell type annotation (achieving up to 96.1% accuracy), cross-dataset integration under heterogeneous antibody panels (successfully aligning over 60,000 cells across 3 distinct multi-omic technologies), and protein-informed cell-cell communication analysis (prioritizing literature-supported interactions with an 81.8% validation rate).

Results

Unifying transcriptome and proteome representations with CAPTAIN

CAPTAIN is a single-cell foundation model pre-trained on large-scale human and mouse datasets comprising co-assayed transcriptome and proteome profiles. It adopts a dual-encoder Transformer architecture, in which RNA and protein modalities are processed independently and integrated via a cross-modal attention mechanism to produce a unified representation of cellular state (Fig. 1 and Methods). To enhance transcriptome modelling, the RNA encoder incorporates a gene knowledge module that encodes prior gene-gene relationships and captures high-order regulatory dependencies. Additionally, CAPTAIN leverages transcriptomic embeddings inherited by its RNA encoder from scGPT, a foundational model pretrained on over 33 million single-cell RNA profiles. This initialization yields robust and transferable transcriptomic representations, enabling broad generalization across diverse biological contexts. The protein encoder is trained jointly with the RNA encoder, and the cross-modal attention module captures fine-grained alignment between transcriptional and proteomic features, resulting in a shared embedding space. CAPTAIN simultaneously optimizes unsupervised representation learning for gene expression reconstruction and supervised metric learning for protein abundance prediction, using a multi-objective training framework (Fig. 1a). This design enables the model to learn intricate transcriptome-proteome relationships and to generate modality-agnostic, biologically grounded representations that support robust transfer to diverse downstream tasks (Fig. 1b).

Despite the vast and rapidly expanding repository of single-cell RNA-seq data, corresponding protein-level measurements remain scarce, and inconsistencies in experimental protocols and antibody nomenclature further hinder effective alignment and joint modelling of the two modalities. We therefore curated and harmonized publicly available co-assay datasets (Fig. 1c, d and Supplementary Data 1–3), resolving naming ambiguities and establishing a unified vocabulary of 382 surface proteins with detailed functional annotation (Fig. 1e and Supplementary Data 4–8). This effort yielded a comprehensive and standardized corpus of co-assayed single-cell transcriptomic and proteomic profiles, termed scT & P-4M, which provides a robust foundation for cross-modal pretraining. scT & P-4M comprises over 4.2 million single cells from both human and mouse, spanning diverse sequencing platforms, tissue origins, and molecular functions across 249 independent samples. The majority of profiles (92.01%) originate from CITE-seq, with the remainder derived from other compatible co-assay platforms. This resource enables precise cross-modal representation learning and makes possible zero-shot inference of previously unseen proteins, while its broad coverage across tissues, species, and platforms ensures that CAPTAIN learns biologically grounded representations with strong generalization capacity.

A key advantage of CAPTAIN's aligned pretraining is its ability to produce generalizable embeddings that can be readily adapted to diverse downstream tasks across both modalities through lightweight fine-tuning. To support this, we developed a flexible fine-tuning

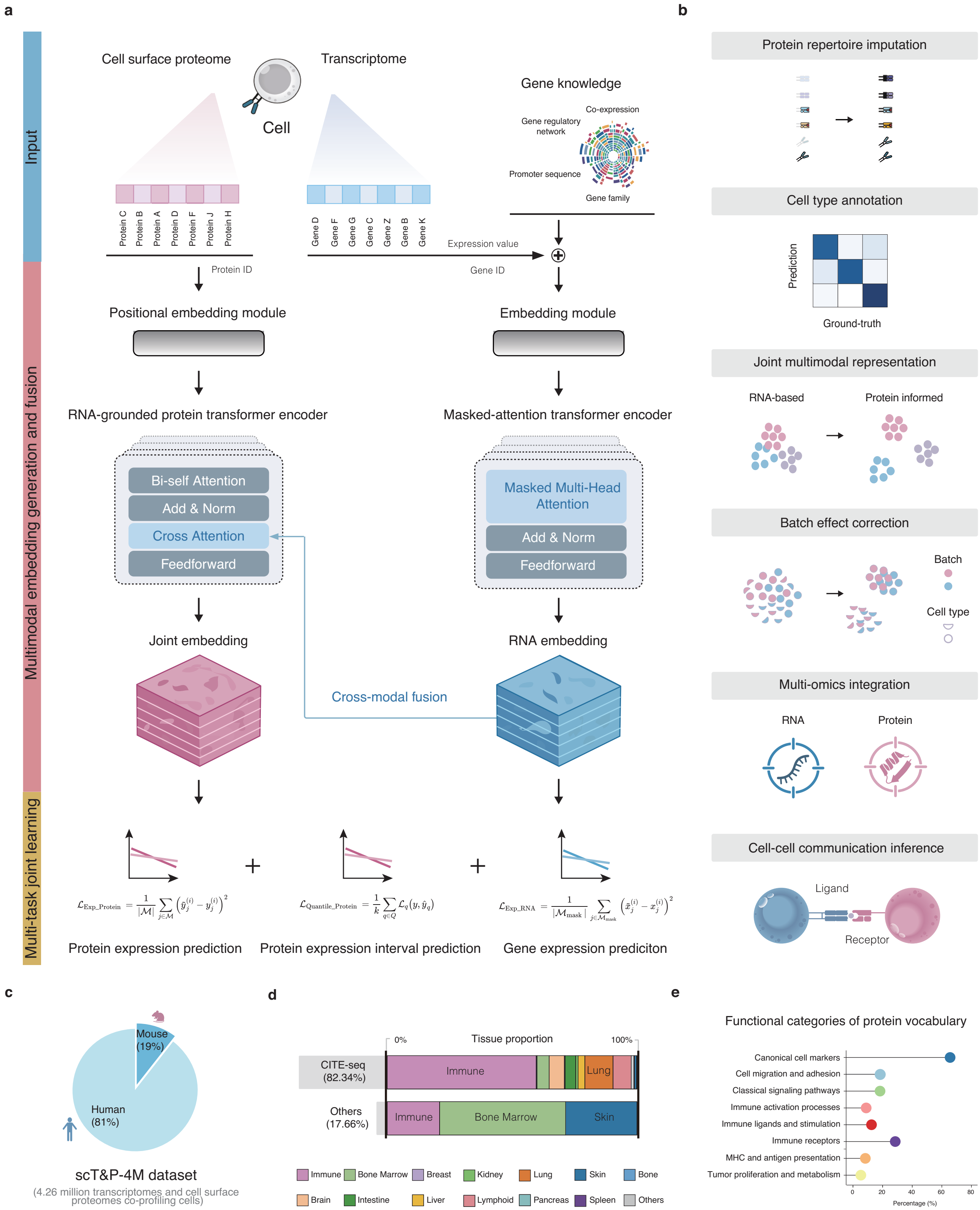

Fig. 1 | Overview of the CAPTAIN model architecture, training strategy, and pretraining corpus. a.CAPTAIN is pre-trained on large-scale co-assayed single-cell RNA-protein data from human and mouse. A Transformer-based architecture learns joint transcriptome-proteome representations with multi-task training (Methods). **b** The pretrained model enables a wide range of downstream single-cell analysis tasks, including protein imputation and expansion, clustering, multi-omic integration, cell type annotation, batch effect correction, and inference of cell-cell communication. **c.** The curated pretraining corpus, scT & P-4M, comprises over 4.2 million co-profiled single cells with matched transcriptomic and surface proteomic measurements from both human and mouse (Supplementary Data 1). **d** The dataset spans more than 13 tissues and organ systems, and 11 disease types, supporting the learning of diverse biological representations (Supplementary Data 2 and 3). **e** After rigorous standardization of nomenclature and functional annotation, 382 distinct surface proteins were used as the proteome vocabulary in CAPTAIN, covering a wide range of molecular functions (Supplementary Data 4–8). Source data are provided as a Source Data file. All schematics in this figure were created by the authors.

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pipeline compatible with a broad spectrum of core applications in single-cell analysis, including protein imputation or expansion from transcriptomic inputs, cell type annotation, batch effect correction, multi-omic integration, and cell-cell communication inference (Fig. 1b and Methods). This modular design enables CAPTAIN to serve as a universal backbone for single-cell analytics, supporting scalable deployment across tissues, modalities, and biological contexts.

CAPTAIN enables zero-shot inference of unmeasured surface proteins

Surface proteins are critical for defining cellular phenotype, immune function, and therapeutic targets[25,32], yet their measurement in single-cell experiments remains both technically demanding and economically costly. Antibody-based modalities such as CITE-seq offer a scalable means to quantify RNA and protein simultaneously, but are constrained by high reagent costs, limited antibody availability, and variable signal-to-noise ratios. As a result, many studies either exclude proteomic profiling altogether or include only a limited subset of proteins. In the scT & P-4M corpus, for instance, protein panel sizes across datasets range from 2 to 210 (mean = 78), more than 4.9-fold smaller than our standardized protein vocabulary. It is therefore critical to assess whether CAPTAIN can accurately infer protein abundance from transcriptomic data. We evaluate this capability under two distinct settings: fine-tuned prediction, where the model is further trained on a subset of paired RNA-protein data from the new dataset and then infers the abundance of all 382 proteins in the standardized protein vocabulary for the remaining cells based solely on their transcriptomic data; and zero-shot prediction. In this latter setting, the model directly infers protein abundance from transcriptomic data alone without any dataset-specific fine-tuning. We clarify that 'zero-shot' here implies that the specific paired expression data were unseen globally (including during pre-training); however, the scope of prediction remains restricted to the 382 proteins present in the standardized protein vocabulary and does not extend to out-of-vocabulary targets.

To this end, we evaluated both the fine-tuned and zero-shot performance of CAPTAIN across four representative CITE-seq datasets spanning different tissues, species, and protein panels. These included a human peripheral blood mononuclear cell (PBMC) dataset with 198 proteins and 20,729 genes[33]; a human mucosa-associated lymphoid tissue (MALT) dataset with 10 proteins and 33,538 genes (10x Genomics); a human monocyte (MNC) dataset with 10 proteins and 29,929 genes[34]; and a mouse PBMC dataset with 80 proteins and 31,056 genes[35]. For each dataset, we held out a subset of cells as the test set and compared CAPTAIN to five state-of-the-art methods (Seurat[33], sciPENN[34], TotalVI[36], scTranslator[37] and scTEL[38]). We quantified prediction accuracy by computing both the Pearson correlation coefficient (PCC) and the root mean squared error (RMSE) between the predicted and observed protein expression.

In the fine-tuned setting, CAPTAIN achieved consistently strong protein prediction performance across all datasets, showing favourable performance relative to established benchmarks including scTranslator, scTEL, Seurat, sciPENN, and TotalVI (Fig. 2a). In the zero-shot setting, CAPTAIN remained highly competitive. While reference-based methods, such as Seurat, may perform strongly when an appropriate reference atlas is available in this domain, it is critical to note a methodological distinction: Seurat employs transfer learning, requiring an annotated reference atlas, whereas CAPTAIN generates predictions solely from its pre-trained knowledge base. As a result, CAPTAIN is particularly suitable for analyzing biological contexts where appropriate reference atlases are unavailable, a setting in which reference-dependent methods are difficult to apply. Even under this more challenging constraint, CAPTAIN's zero-shot performance was comparable to, and in certain datasets (e.g., human MALT and mouse PBMC), modestly exceeded that of Seurat. Furthermore, when experimental protocols are aligned (comparing Seurat against fine-tuned CAPTAIN), our model consistently achieves higher prediction accuracy under matched experimental protocols, demonstrating that CAPTAIN leverages available supervision more effectively than Seurat's transfer approach. Notably, compared to the recent scTranslator[37], CAPTAIN shows improved coverage and stability in the zero-shot context, partly reflecting differences in output formulation and gene-protein mapping strategies. In the mouse PBMC dataset, evaluation was performed using a benchmark protein panel consisting of 80 proteins. CAPTAIN generated predictions for all proteins in this panel, whereas scTEL generated predictions for only 19 proteins and only one predicted protein from scTranslator overlapped with the evaluation panel. Consequently, the number of proteins that could be directly compared across methods was extremely small, making a systematic cross-method comparison infeasible; the available results are therefore summarized in Supplementary Data 9. Predictive performance on these few comparable proteins was also weak, further limiting meaningful evaluation for these methods. In contrast, CAPTAIN consistently generated predictions across the entire protein panel. These results suggest that CAPTAIN provides robust estimations (Supplementary Data 9), making it a practical alternative when paired protein data or computational resources are limited. While Pearson correlation confirms that CAPTAIN captures the correct biological trends, strictly linear metrics can be insensitive to systematic shifts. To rigorously validate the benchmarking results, we visualized CAPTAIN's predictions for key immune markers in human PBMCs. The predicted and observed expression values of CAPTAIN showed strong concordance, with tight clustering along the diagonal. Notably, CAPTAIN preserves this structural consistency even in zero-shot scenarios (Fig. 2b and Supplementary Fig. 1), whereas existing methods exhibit divergent distributions and substantial error variance.

To further highlight CAPTAIN's most prominent zero-shot capability, we then present the predicted expression patterns for proteins CD21, CD79, CD4, and CD94 under the zero-shot setting. Notably, these proteins exist in our protein vocabulary but were entirely unmeasured in the dataset. The predicted expression patterns of these proteins closely matched the known canonical markers of their respective cell types (e.g., B cells, $CD4^+$ T cells, NK cells), despite the absence of any paired protein measurements during training (Fig. 2c). These findings highlight CAPTAIN's ability to infer surface protein expression in a zero-shot setting for proteins unmeasured in new datasets, thereby expanding the effective proteomic feature space from transcriptomic inputs. This capability reflects the advantages of protein-aware pretraining in multimodal representation learning and reflects the advantages of CAPTAIN's biologically grounded architecture and extensive multi-omic pretraining.

Robust cell type annotation across modalities and resolutions

Leveraging its pretraining on large-scale multi-omic data, CAPTAIN enables accurate cell type annotation than models trained solely on scRNA-seq data. To fine-tune CAPTAIN for this task, a neural network classifier was added to process cell embeddings from the protein encoder and output categorical cell type predictions. This composite model was trained end-to-end using cross-entropy loss on a reference dataset with expert-curated annotations and then applied to a separate query partition. CAPTAIN was evaluated across multiple single-cell transcriptomic and multi-omic CITE-seq datasets (Supplementary Data 10). In the human PBMC CITE-seq dataset, UMAP visualizations coloured by reference annotations (Fig. 3a) and by CAPTAIN's predicted labels (Fig. 3b) showed strong concordance, yielding an overall classification accuracy of 96.1%. The corresponding confusion matrix (Fig. 3c) indicated high precision (above 90%) across most cell types. Slight reductions in performance were observed for rare cell populations in the reference set, including $CD8^+$ effector memory T (Tem)

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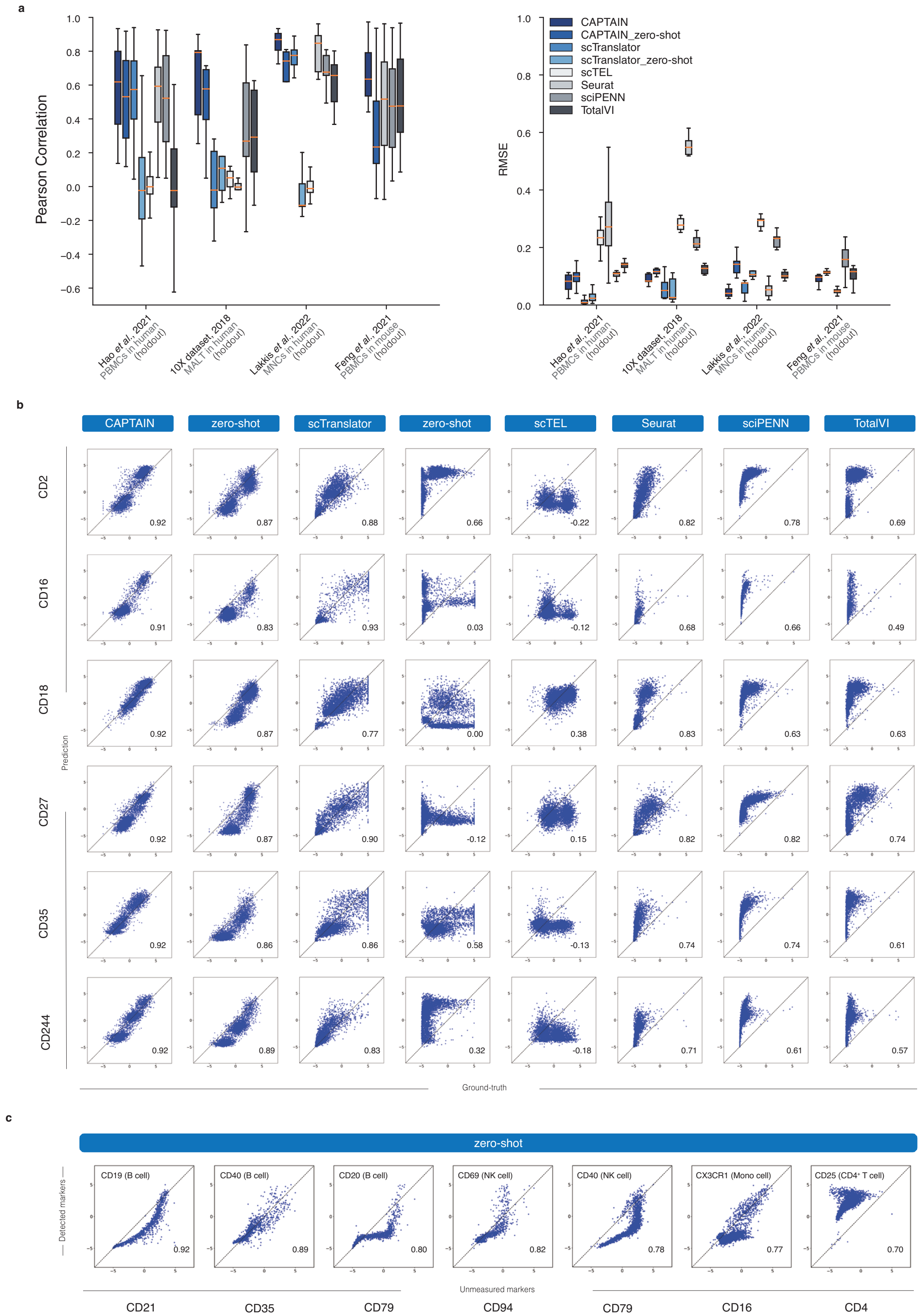


cells, mucosa-associated invariant T (MAIT) cells, and plasmacytoid dendritic cells (pDCs).

To evaluate the generalization capability and robustness of CAPTAIN for cell type annotation across diverse modalities and datasets, we conducted a systematic benchmarking study using five scRNA-seq and five CITE-seq cohorts. This evaluation encompassed both "held-in" datasets from the pre-training corpus and "held-out" datasets to rigorously test the model's performance on unseen data. Applying a five-fold cross-validation strategy, we calculated the mean performance across all folds for five key evaluation metrics: *Accuracy*, *Precision*, *Recall*, *Macro-F1*, and an Overall score. As illustrated in Fig. 3d, CAPTAIN (fine-tuned) consistently outperformed competing methods in both scRNA-seq and CITE-seq modalities. In the scRNA-seq benchmarks, CAPTAIN maintained a superior ranking compared to

Fig. 2 | CAPTAIN predicts surface protein expression from transcriptomic inputs in fine-tuned and zero-shot settings. a Prediction performance was evaluated by comparing CAPTAIN with Seurat, sciPENN, TotalVI, scTranslator and scTEL across four CITE-seq datasets. Held-out test sets excluded from fine-tuning were used to evaluate both fine-tuned and zero-shot configurations, with Pearson correlation coefficient (PCC, left panel) and root-mean-square error (RMSE, right panel) as evaluation metrics. In the mouse PBMC dataset, evaluation was performed using a benchmark protein panel consisting of 80 proteins. CAPTAIN generated predictions for all proteins in this panel, whereas scTEL generated predictions for only 19 proteins and only one predicted protein from scTranslator overlapped with the evaluation panel. Because the number of evaluable proteins for these methods was extremely small, meaningful cross-method comparison was not feasible, and their results are summarized in Supplementary Data 9. Statistical analysis using two-sided Wilcoxon signed-rank tests comparing CAPTAIN's Pearson correlation coefficients against the second-best method yielded *p*-values of 3.35×10^{-26}, 7.81×10^{-3}, 5.46×10^{-2}, and 1.17×10^{-13} across the four datasets, respectively. Statistics in the box plots were derived from independent protein-wise prediction metrics across the four datasets, where the precise number of distinct proteins (*n*) evaluated from left to right is $n = 198$, 10, 10, and 80, respectively. In the box plots, the centre line indicates the median, the bounds of box indicate the 25th and 75th percentiles, and the whiskers extend to the minima and maxima values. **b** Marker-level prediction accuracy was examined in the human PBMC dataset by comparing predicted and measured expression values for canonical immune proteins. Predictions from CAPTAIN aligned closely with measured values in both fine-tuned and zero-shot settings, whereas other methods showed greater variance and systematic deviation from the identity line. Protein counts were standardized to have zero mean and unit variance during preprocessing. For visualization, both predicted and measured values are displayed using a fixed axis range of [−5, 5] to ensure consistent scaling across methods. No data points were clipped or removed during visualization. **c** Unmeasured proteins that were completely excluded from fine-tuning were predicted by CAPTAIN and evaluated by comparison to the expression of known proxy markers representing the same cell types. Source data are provided as a Source Data file.

established models such as scGPT and Seurat. Similarly, in the CITE-seq context, which requires the integration of surface protein and gene expression data—CAPTAIN demonstrated enhanced stability and higher mean scores across all metrics. These results highlight CAPTAIN's ability to effectively leverage its pre-trained representations to achieve state-of-the-art accuracy and robustness in cell type identification (Detailed numerical results are provided in Supplementary Data 10).

Next, we sought to assess CAPTAIN's capacity for fine-grained annotation by applying it to a CITE-seq dataset of bone marrow mononuclear cells with literature-defined labels for 13 T cell subtypes[39]. UMAP projections based on reference annotations revealed distinct spatial segregation between $CD4^+$ and $CD8^+$ subsets, with CD4 subtypes concentrated in the upper right and CD8 subtypes in the lower left (Fig. 3e). Protein expression patterns recapitulated this separation, whereas gene expression alone was insufficient to clearly distinguish the lineages, highlighting the value of surface protein information for resolving closely related cell states (Fig. 3g,h). CAPTAIN accurately predicted the corresponding marker proteins, reinforcing its ability to infer proteomic distinctions from transcriptomic input. It achieved a *Macro-F1* score of 0.73 across all T cell subtypes, substantially outperforming Seurat (0.61) and scGPT (0.04) (Fig. 3f and Supplementary Fig. 2-3). scGPT misclassified all T cells as $CD4^+$ T naive, while Seurat only partially distinguished $CD4^+$ and $CD8^+$ groups. These results highlight CAPTAIN's strength in fine-grained cell type annotation, enabled by its large-scale multi-omic pretraining and incorporation of biological priors, which together yield more discriminative and generalizable representations than unimodal RNA-based models.

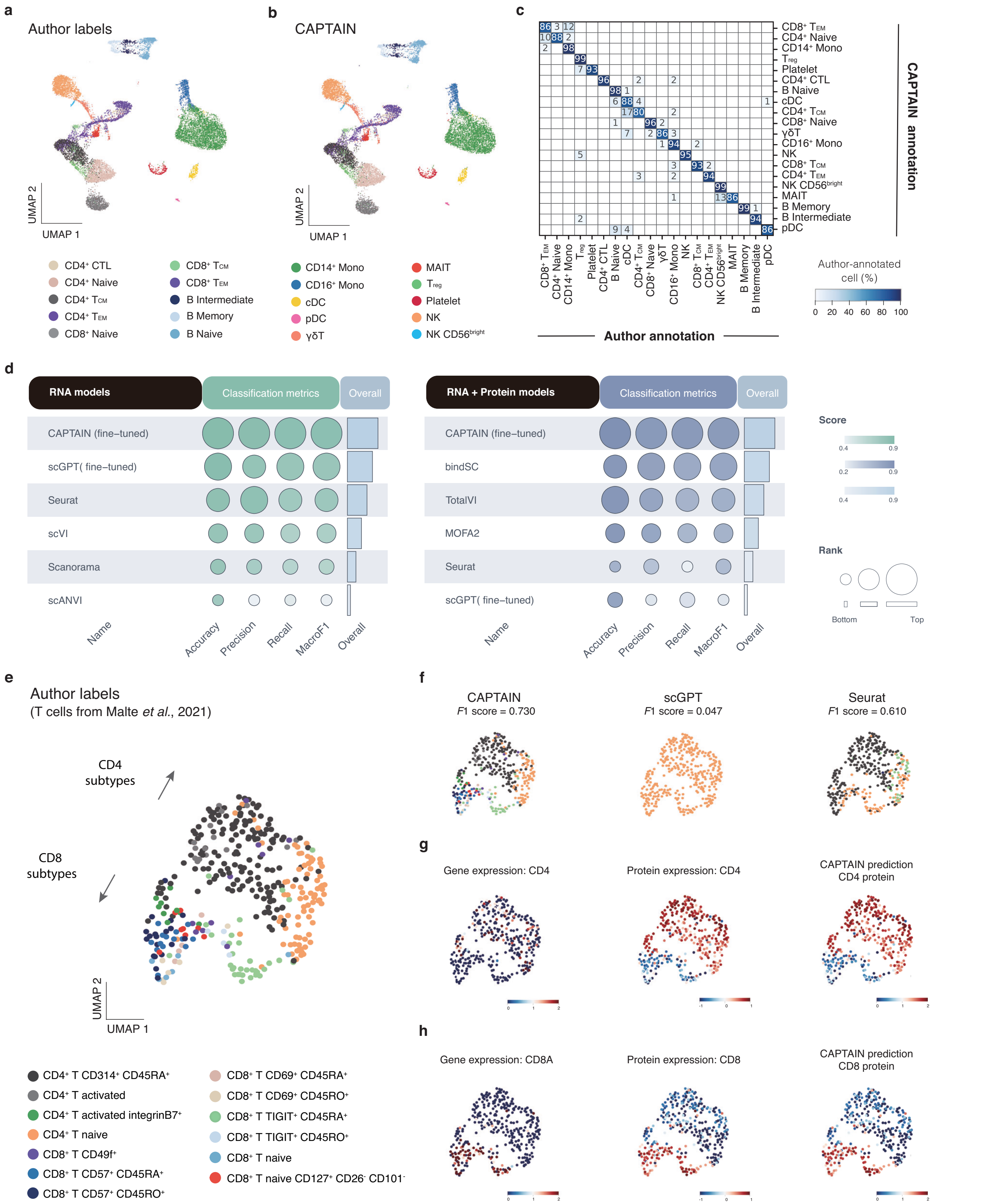
Biologically meaningful integration across batches, modalities and platforms

Integrating multi-batch single-cell sequencing data remains a persistent challenge, necessitating the removal of spurious technical variations while preserving genuine biological heterogeneity. To address this, we fine-tuned CAPTAIN (see Methods) to learn unified representations across diverse technical and biological conditions.

We first benchmarked CAPTAIN against six widely used integration methods, including Seurat[33], Harmony[40], scANVI[41], scVI[42], Scanorama[43], and scGPT[12], across three multi-batch scRNA-seq datasets: PBMCs from 8 healthy donors[33], PBMCs from 2 SARS-CoV-2 infected individuals[44], and BMMCs from 13 healthy donors[39] (Supplementary Data 11). The averaged performance across these datasets is presented in Fig. 4a (left). Subsequently, to further enhance biological representation, we expanded the input space to incorporate biologically driven protein and RNA modalities. We evaluated CAPTAIN's performance in integrating single-cell multi-omics datasets, which present elevated complexity due to molecular heterogeneity. By leveraging multi-modal pre-training, CAPTAIN learns unified embeddings directly from diverse input types. We compared CAPTAIN against six prominent multi-omics integration methods (Seurat[33], Scanpy[45], MOFA2[46], TotalVI[36], bindSC[47], and scGPT[12]) using a paired RNA-protein COVID-19 CITE-seq dataset and heterogeneous datasets generated by CITE-seq, ECCITE-seq, and TEA-seq (Supplementary Data 12; averaged results: Fig. 4a right). The results demonstrate that CAPTAIN (fine-tuned) consistently achieves strong performance in both scRNA-seq and CITE-seq integration tasks. As illustrated in the ranking metrics, CAPTAIN ranks among the top-performing methods, reflecting a favourable balance between batch correction and biological conservation.

On the PBMC Healthy dataset, CAPTAIN effectively aligned cells across batches while preserving clear boundaries between major immune cell types (Fig. 4b). Quantitatively, CAPTAIN achieved the highest Biological Conservation score (*AvgBIO*), which summarizes three metrics: NMI_{cell}, ARI_{cell}, and ASW_{cell} (Supplementary Data 11). Decomposition of *AvgBIO* revealed that the largest difference between CAPTAIN and the other methods arises from the ARI_{cell} metric. In particular, CAPTAIN shows an average ARI_{cell} advantage of approximately 0.216 over scVI and scANVI (Supplementary Data 11; Fig. 4b). Because ARI_{cell} measures the agreement between clustering structure and annotated cell-type labels, it is particularly sensitive to partial mixing between closely related subtypes. As a result, embeddings may appear well separated at a coarse cell-type level while still exhibiting reduced ARI_{cell} values when subtype-level boundaries are not well preserved. To illustrate this effect more directly, we visualized the integrated embeddings using PCA. Compared with UMAP, which emphasizes local neighbourhood structure, PCA better reflects the global organization of the latent space and more clearly reveals subtype-level overlap. In the embeddings generated by scVI and scANVI, closely related T-cell subtypes (e.g., $CD4^+$ Naive and $CD8^+$ Naive) show increased overlap, whereas CAPTAIN maintains clearer separation between these populations (Fig. 4b). Together, the quantitative metric decomposition and PCA visualization explain why scVI and scANVI can appear visually well separated at a coarse level while still achieving lower ARI_{cell} and consequently lower *AvgBIO* scores.

Among various batch-effect correction tasks, cross-platform integration remains one of the most challenging problems. Differences in sequencing time, sample origin, library preparation protocols, and operator variability introduce substantial technical noise and systematic biases into the data. To rigorously evaluate CAPTAIN's performance under such complex conditions, we applied it to heterogeneous single-cell datasets generated using three multi-omic technologies: CITE-seq, ECCITE-seq, and TEA-seq. Although these datasets were originally produced independently, they were jointly analyzed within a unified embedding space comprising over 60,000



cells (Fig. 4d)[33,48]. This design spans the full technical spectrum of multimodal sequencing, from CITE-seq (measuring RNA and protein), to ECCITE-seq (additionally capturing TCR and CRISPR readouts), and further to TEA-seq (integrating RNA, protein, and chromatin accessibility).

Given the profound technical heterogeneity among these protocols, particularly the distinct data distribution of TEA-seq, we acknowledge that standard aggregate metrics may mask residual batch effects. For instance, while initial evaluations yielded high *AvgBATCH* scores, the corresponding biological conservation was disproportionately lower (*AvgBIO* = 0.568), suggesting that a more granular assessment was required. To this end, we expanded our evaluation framework to include three complementary integration metrics, ARI_{batch}, iLISI, and kBET scores and extended our benchmark

Fig. 3 | CAPTAIN enables accurate and fine-grained cell type annotation across diverse single-cell datasets. **a** UMAP visualization of a human PBMC CITE-seq dataset, with cells coloured by author-provided reference annotations. **b**. Same UMAP projection as in (a), with cells coloured by CAPTAIN-predicted labels, showing strong concordance with the reference. **c**. Confusion matrix comparing CAPTAIN's predictions to the reference annotations in the PBMC dataset. Each column corresponds to a reference cell type, and each row to a predicted label; values represent the percentage of cells from each true population assigned to each predicted class. **d** Comparison of CAPTAIN (fine-tuned) and competing methods for cell type annotation averaged across multiple scRNA-seq and CITE-seq datasets. The visualization displays five evaluation metrics, where circle size is linearly scaled according to the rank order of the methods, and colour saturation (green for scRNA-seq; blue for CITE-seq) reflects the overall performance ranking (1–6). Detailed numerical results are provided in Supplementary Data 10. **e** UMAP visualization of a bone marrow mononuclear cell (BMMC) CITE-seq dataset, coloured by fine-grained T cell subtype labels from the original study. **f** UMAP visualizations of CAPTAIN, scGPT, and Seurat predictions for the same dataset, with *Macro-F1* scores reported above each panel. Marker-level expression visualizations for CD4 (**g**) and CD8A (**h**) in the BMMC dataset. Each panel shows ground-truth gene expression, ground-truth protein expression, and CAPTAIN-predicted protein expression. Protein-level signals offer clearer separation between $CD4^+$ and $CD8^+$ T cells than gene expression, and CAPTAIN's predictions closely match true protein distributions. Source data are provided as a Source Data file.

comparison to include totalVI (Supplementary Data 19). Notably, the *AvgBATCH* score displayed in Fig. 4d retains its original calculation to maintain consistency with our primary evaluation framework across the main figures, and does not incorporate these newly added metrics. As shown in Supplementary Data 19, the relatively low iLISI and kBET scores observed across all methods highlight the inherent difficulty of this cross-platform integration task. Within this context, CAPTAIN achieves strong overall performance and ranks among the top-performing methods, reflecting a favourable balance between mitigating technical variation and preserving biologically meaningful structure.

CAPTAIN enables protein-informed cell-cell communication inference

Cell-cell communication orchestrates biological processes by transmitting signals through ligand-receptor interactions. These signalling events underlie tissue homeostasis, immune coordination, and therapeutic response. Although surface proteins directly mediate such interactions, most existing computational methods infer intercellular communication solely from transcriptomic data, owing to the widespread availability of scRNA-seq. However, the exclusion of protein information yields an incomplete view of the cellular signalling machinery, particularly the receptor landscape that determines cellular responsiveness[49]. In contrast, protein-informed inference offers a more intrinsic and biologically grounded representation of intercellular communication. To evaluate whether CAPTAIN can overcome the limitations of transcriptome-only models, we developed a hybrid framework that integrates ligand expression from the transcriptome with receptor abundance imputed from the proteome. CAPTAIN first predicts a comprehensive surface protein profile from RNA data using either zero-shot inference or fine-tuning on paired multi-omic datasets. Ligand-receptor interactions are then inferred by combining transcript-derived ligand levels with predicted receptor abundance, and their statistical significance is assessed through permutation testing. This framework enables the reconstruction of signalling networks that more closely reflect underlying biological processes (Fig. 5a and Methods).

We benchmarked CAPTAIN on a CITE-seq PBMC dataset[33] containing paired RNA and surface protein profiles, comparing its performance against five widely used methods: NATMI[50], Connectome[51], CellChat[52], CellphoneDB[53] and SingleCellSignalR[54]. Focusing on $CD4^+$ naive T cells, which play a central role in coordinating immune responses[55], we evaluated predicted signalling to NK cells (Fig. 5b and Supplementary Fig. 4). Given that existing methods rely on raw multi-omics data, whereas CAPTAIN explicitly performs protein imputation to recover events missed by RNA-only approaches, we utilized literature curation as an external biological reference to assess the plausibility of these novel signals. CAPTAIN prioritized 22 ligand-receptor interactions, of which 18 (81.8%) were supported by existing literature. In contrast, the strongest alternative method recovered 6 interactions, while several others detected none (Supplementary Data 13).

To address the lack of ground truth in single-cell communication and provide additional functional validation, we further assessed whether the predicted interactions induced expected downstream transcriptional changes in receiver cells[56]. We stratified receiver cells (NK cells) into "Strong" and "Weak" communication groups based on inferred interaction strength. As shown in Fig. 5d, NicheNet-prioritized target genes associated with the top-predicted ligand-receptor pairs were significantly upregulated ($p < 0.001$, two-sided Wilcoxon rank-sum test) in the "Strong" communication group. These results indicate that the inferred interactions are associated with expected downstream transcriptional responses in receiver cells, supporting their biological relevance. GO enrichment analysis further revealed a significant overrepresentation of immune-related functions, including MHC protein binding and transmembrane receptor kinase activity.

To further assess the zero-shot capability of CAPTAIN, as well as the robustness of its inference strategy, we utilized another independent scRNA-seq PBMC dataset[44] where surface proteins were not experimentally measured (Supplementary Fig. 4). This setting highlights a distinguishing capability of CAPTAIN, as it is the only method capable of performing protein-informed inference through zero-shot imputation. This analytical distinction necessitated a shift in our evaluation strategy: rather than benchmarking for sensitivity to novel signals (as with the CITE-seq data), our objective here was to assess the robustness of CAPTAIN with respect to the prevailing biological consensus. We therefore evaluated concordance with established, transcriptome-only methods. Of the 16 interactions predicted between $CD4^+$ T cells and dendritic cells (DCs), 15 were also identified by at least one established method, demonstrating strong concordance (Fig. 5c, middle panel). We replicated our functional validation on this dataset and observed a consistent pattern: receiver cells (DCs) in the "Strong" signalling group exhibited significantly higher expression of downstream target genes compared to those in the "Weak" group (Fig. 5d, bottom panel). These results indicate that CAPTAIN captures key features of the consensus landscape of cellular communication.

To investigate how intercellular communication changes with disease progression, we applied CAPTAIN to a multi-sample COVID-19 dataset[57]. The analysis revealed a progressive increase in platelet-to-monocyte signalling with advancing disease severity, consistent with previous findings[58] (Fig. 5e and Supplementary Fig. 5). Notably, CAPTAIN's protein-centric inference pinpointed the disease-dependent activation of the pro-inflammatory S100A9-CD36 axis, which remained inactive in healthy and asymptomatic individuals but became increasingly prominent in mild to severe cases. This finding illustrates a capability enabled by CAPTAIN's protein-informed inference that is difficult to achieve reliably with transcriptome-only ligand-receptor inference tools under incomplete protein measurement. (e.g., CellChat, CellphoneDB). These existing methods rely strictly on observed input profiles and lack the capacity to impute unmeasured proteins; consequently, they may fail to detect interactions driven by proteins absent from the antibody panel or those decoupled from their mRNA abundance. Indeed, prior studies indicate that the pathogenic role of

Fig. 4 | CAPTAIN enables accurate multi-batch and multi-omic single-cell data integration. a Comparison of CAPTAIN (fine-tuned) and competing methods for multi-batch and multi-omic data integration averaged across multiple scRNA-seq and CITE-seq datasets. The visualization displays eight evaluation metrics grouped by Batch Correction and Biological Conservation, where symbol size (bars and circles) is linearly scaled according to the rank order of the methods, and colour saturation (grey for Overall; teal for Batch correction; blue for Biological conservation) reflects the performance ranking (1–7). Detailed numerical results are provided in Supplementary Data 11 and 12. **b** Quantitative comparison of integration metrics and PCA visualization of integrated embeddings on the PBMC Healthy dataset. Left: Line plots summarizing Biological Conservation metrics across integration methods, including *AvgBIO* and its component metrics (ARI_{cell}, NMI_{cell}, and ASW_{cell}). The largest performance gap between CAPTAIN and scVI/scANVI is observed in ARI_{cell} (Supplementary Data 11). Right: PCA projections (PC1 vs PC2) of the integrated embeddings produced by each method. The upper panels show cells coloured by batch to assess batch mixing, while the lower panels show cells coloured by cell-type annotations. Closely related T-cell subtypes (e.g., CD4 + Naive and CD8 + Naive) exhibit increased overlap in embeddings generated by scVI and scANVI, whereas CAPTAIN maintains clearer separation between these subpopulations. **c** UMAP embeddings of the BMMC CITE-seq dataset, generated by CAPTAIN, scGPT, TotalVI, and Scanpy. Cells are coloured by fine-grained cell subtype annotations to visualize the quality of integrated cell clustering. **d** UMAP embeddings from CAPTAIN and scGPT for heterogeneous datasets spanning three multi-omic technologies: CITE-seq, ECCITE-seq, and TEA-seq. For each method, the left panel shows cells coloured by technology platform to assess batch alignment, and the right panel shows cell type labels to assess biological structure preservation. *AvgBATCH* and *AvgBIO* scores are displayed for each method. Source data are provided as a Source Data file.

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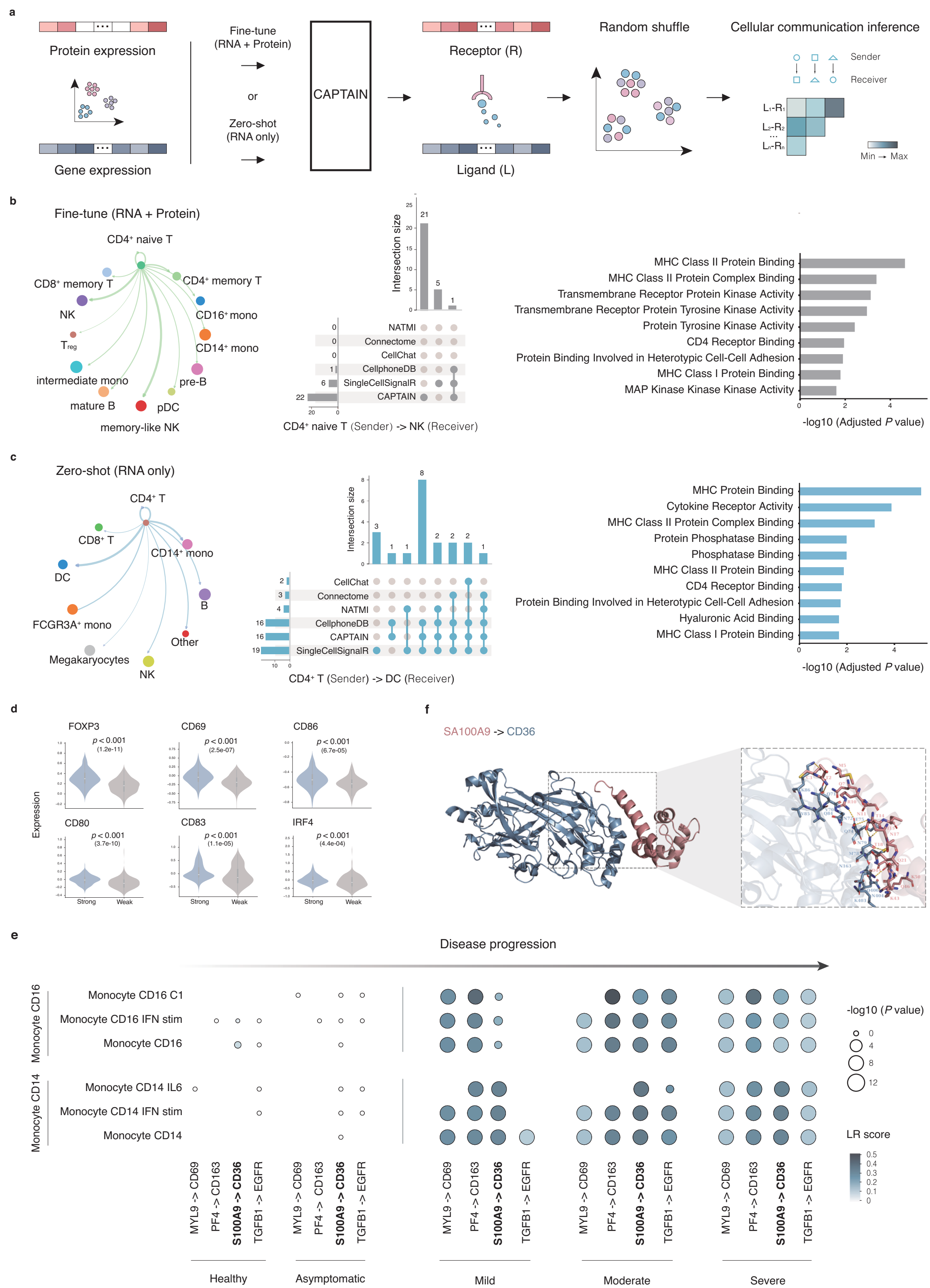

CD36 in this context is driven by post-transcriptional regulation and enhanced surface presentation rather than increased gene expression. To further examine the biophysical plausibility of this inferred interaction, we performed protein-protein docking simulations using HDOCK (Supplementary Data 20). The analysis yielded a docking configuration with favourable scores and geometric compatibility, providing structural plausibility for the proposed interaction (Fig. 5f). While transcriptome-only tools may fail to identify this interaction due to low RNA correlation under conditions of incomplete protein measurement, CAPTAIN leverages its cross-modal framework to computationally complete the proteomic landscape. Thus, the ability to capture this biologically plausible and structurally feasible axis highlights the potential value of protein-informed frameworks for generating testable hypotheses in complex human diseases.

Fig. 5 | CAPTAIN infers biologically meaningful cell-cell communication and generates hypotheses regarding potential disease-specific signalling dynamics. a Schematic of the CAPTAIN-based communication inference workflow. Ligand expression is extracted from the transcriptome, and receptor abundance is reconstructed using CAPTAIN-predicted protein profiles. Statistical significance of interactions is assessed via permutation testing. **b** Benchmarking on a CITE-seq PBMC dataset. Left: Circos plot showing inferred signalling from CD4⁺ naive T cells. Middle: UpSet plot comparing interactions identified by CAPTAIN against five widely used methods (NATMI, Connectome, CellChat, CellphoneDB, SingleCellSignalR). Right: GO enrichment analysis highlights immune-relevant molecular functions of CAPTAIN-inferred interactions. **c** Zero-shot inference on an independent scRNA-seq PBMC dataset. Left: Network plot of inferred CD4⁺ T - DC interactions. Right: Bar plot showing the intersection of predictions with other methods. **d** Validation of inferred interactions via downstream target expression. Receiver cells were stratified into 'Strong' and 'Weak' groups based on CAPTAIN-predicted interaction scores. Violin plots display the expression of NicheNet-prioritized target genes for top ligand-receptor pairs in the top panels (Dataset 1: CD4⁺ naive T and NK cells) and the bottom panels (Dataset 2: CD4⁺ T and DC cells). p values were calculated using a two-sided Wilcoxon rank-sum test (exact p-values: FOXP3, $p = 1.2 \times 10^{-11}$; CD69, $p = 2.5 \times 10^{-7}$; CD86, $p = 6.7 \times 10^{-5}$; CD80, $p = 3.7 \times 10^{-10}$; CD83, $p = 1.1 \times 10^{-5}$; IRF4, $p = 4.4 \times 10^{-4}$). Statistics and p-values in the violin plots were derived from independent cell-wise expression values, where the precise number of distinct cells (n) evaluated for both the Strong and Weak groups is $n = 100$ per group for each gene. For the embedded box plots within the violin plots, the centre line indicates the median, the bounds of box represent the 25th and 75th percentiles, and the whiskers extend to the minima and maxima values. **e** Application to a multi-sample COVID-19 dataset. Circos plots show platelet-to-monocyte communication across increasing disease severity (Healthy, Asymptomatic, Mild, Moderate, Severe), highlighting progressive activation of the S100A9-CD36 axis. **f** Structural validation of the S100A9-CD36 interaction. The top-ranked protein-protein docking model generated by HDOCK visualizes the physical binding between S100A9 (salmon) and CD36 (blue). The inset highlights the binding interface and key residue contacts, supporting the biophysical feasibility of this CAPTAIN-imputed interaction. Source data are provided as a Source Data file.

Discussion

In this study, we introduce CAPTAIN, a multimodal foundation model trained on scT & P-4M, a massive corpus of over 4.2 million co-assayed transcriptomic and proteomic single-cell profiles. CAPTAIN establishes a unified framework for representing cellular state by integrating transcriptional and surface protein features through a cross-attention Transformer architecture. This design supports the inference of surface protein abundance, fine-resolution cell type annotation, data integration, and cell-cell communication analysis. Through multi-task pretraining, CAPTAIN learns transferable representations that can be applied across multiple modalities, tissues, species, and experimental settings.

While CAPTAIN builds upon existing transformer-based frameworks for single-cell analysis, its contribution lies in system-level integration of large-scale multimodal data, standardized protein vocabularies, and protein-aware pretraining rather than in architectural novelty. A central strength of CAPTAIN lies in its ability to infer surface protein expression directly from transcriptomic inputs. Across multiple benchmarks, CAPTAIN shows strong protein inference performance in both fine-tuned and zero-shot settings, with reduced sensitivity to dataset shifts compared with several existing methods (significantly outperforming 5 state-of-the-art methods in prediction accuracy, $p < 0.05$). This robustness extends beyond standard immune profiles; our expanded benchmarking across kidney, spleen, and breast tissues, along with evaluations on a multi-tissue atlas spanning 49 cell types (Supplementary Note 11 and Supplementary Fig. 13), supports the applicability of CAPTAIN to non-immune biological contexts represented in the current benchmarking datasets. Such capabilities enable downstream analyses that require multi-modal resolution even in the absence of complete measurements. For example, CAPTAIN can reconstruct intercellular communication networks by integrating transcriptome-derived ligand expression with imputed receptor abundance. In our benchmarks, this protein-informed approach prioritized a larger set of literature-supported ligand-receptor interactions relative to baseline methods. Specifically, CAPTAIN successfully prioritized 22 interactions with an 81.8% literature validation rate (18 out of 22 pairs), representing a 3-fold improvement over the strongest alternative RNA-only method, which recovered only 6 pairs. This protein-informed inference enables further exploration of transcript-protein relationships, including cases of post-transcriptional decoupling.

CAPTAIN is pre-trained using a standardized vocabulary of 382 surface proteins curated from our scT & P-4M corpus. This unified panel enables consistent modelling across studies and allows CAPTAIN to transcend the sparse and heterogeneous antibody panels typical of current datasets. By predicting this full vocabulary, CAPTAIN substantially expands the effective proteomic feature space by up to 4.9-fold (from an average of 78 measured proteins to the full 382 targets) relative to typical CITE-seq antibody panels. In addition, the scT & P-4M corpus provides a foundation for cross-modal learning, serving as a basis for constructing protein-aware single-cell representations. Although the current version focuses on RNA and surface proteins, future extensions incorporating intracellular proteins, chromatin accessibility, or metabolic data will further enhance its versatility.

Crucially, CAPTAIN differs from earlier transformer-based approaches in its emphasis on protein-aware pretraining, in which surface proteins are explicitly co-modelled during large-scale training rather than introduced only as task-specific outputs. This design is associated with more stable cross-dataset and cross-task transfer compared with dataset-specific fine-tuning strategies. Quantitatively, this translates to state-of-the-art performance in downstream tasks: CAPTAIN achieved a 96.1% overall classification accuracy in cell type annotation, effectively resolving fine-grained T-cell subtypes with a Macro-F1 score of 0.73 (compared to 0.61 for Seurat). Furthermore, CAPTAIN demonstrated exceptional multi-omic integration capabilities by successfully aligning over 60,000 cells across 3 highly heterogeneous sequencing platforms (CITE-seq, ECCITE-seq, and TEA-seq) while preserving biological variance. Despite these advancements, we acknowledge limitations inherent to the current landscape of single-cell multimodal modelling. Our stratified analyses reveal systematic performance differences between high- and low-frequency proteins, reflecting the long-tail distribution and sparsity of existing multimodal datasets (Supplementary Data 15). This effect is closely tied to protein panel design: current CITE-seq and REAP-seq experiments preferentially include a limited set of surface markers, most prominently CD-associated proteins for immunophenotyping, while many non-CD surface proteins and signalling receptors remain sparsely measured or entirely absent. As a consequence, protein prediction performance for underrepresented proteins may appear optimistic, as such proteins are often evaluated only in datasets where they are measured by design, while being absent from both training and testing in other cellular contexts (Supplementary Fig. 7, Supplementary Notes 5 and 6). This limitation reflects structural constraints of prevailing experimental designs rather than a modelling artifact specific to CAPTAIN. Although leveraging the scale of the scT & P-4M dataset and a standardized vocabulary of 382 surface proteins helps aggregate sparse signals and stabilize representation learning, uneven protein coverage remains a fundamental challenge. Importantly, we reiterate that CAPTAIN's zero-shot prediction capabilities are inherently bounded by this predefined protein vocabulary and the biological coverage of current datasets. Accordingly, zero-shot and fine-tuned protein predictions should be interpreted with appropriate caution for sparsely measured proteins and underrepresented tissues, and future progress will depend on broader and more balanced protein panels, as well as complementary data sources that can provide informative priors for proteins that are currently underrepresented in single-cell multimodal assays.

Similarly, the generalization of multimodal models across anatomical sites is influenced by data availability. While our expanded evaluation indicates that CAPTAIN can be applied to several non-immune tissues (including kidney, spleen, and breast tissues), the global multimodal data landscape remains heavily skewed toward immune-related tissues (e.g., PBMC and BMMC). Consequently, generalization to all cell types, particularly rare populations in underrepresented organs, is inherently constrained by the scarcity of paired training measurements. Accordingly, although CAPTAIN generalizes beyond immune-focused benchmarks, its performance and coverage remain influenced by the tissue and protein representation present in the pretraining corpus, reflecting a shared limitation of current multimodal single-cell resources that arises from broader structural constraints in prevailing multimodal experimental designs. We anticipate that future performance gains will therefore be data-centric, driven by the availability of more diverse tissue atlases and balanced protein panels as the field expands beyond immune-centric datasets.

Furthermore, inferring cell-cell communication from dissociated single-cell data entails unavoidable assumptions. While we support our predictions with functional validation of downstream targets and complementary molecular docking analyses, these inferences rely on static ligand-receptor databases and currently lack physical spatial constraints. Consequently, predictions for spatially dependent signalling should be interpreted as potential interaction propensity rather than physical contact, a limitation that future integration with spatial transcriptomics data aims to resolve.

Taken together, CAPTAIN provides a foundation model that supports multimodal single-cell analysis under current data availability, enabling data integration and protein-aware inference across diverse datasets. As multimodal resources continue to expand, such models may facilitate increasingly comprehensive analyses of cellular states and interactions. With continued growth in dataset depth, resolution, and diversity, CAPTAIN may enable deeper insights into cell identity, state transitions, and intercellular dynamics.

Methods

Pre-training data collection and preprocessing

To generate a high-quality pre-training dataset for the CAPTAIN foundation model, we systematically collated and processed publicly available co-assayed single-cell RNA and protein data to construct a scT & P-4M database (Supplementary Note 1). This process comprised three main stages: data acquisition and selection; gene and protein symbol unification; and cell-level quality control and integration. This harmonisation procedure was essential to achieve structural consistency and comparability between the RNA and protein modalities across diverse projects and platforms.

Data acquisition and selection. We performed a comprehensive screen of recent studies, using public repositories including the Gene Expression Omnibus (GEO)[59], ArrayExpress[60] and the official 10x Genomics website. Inclusion criteria were as follows: (i) the dataset had to contain both RNA and protein modalities; (ii) raw count matrices had to be available in a standard format (for example, CSV, H5AD, RDS, or FASTA files subsequently processed using CellRanger); (iii) complete protein annotation was required, with a preference for datasets containing cell-type annotations; and (iv) sample batch and sequencing platform information had to be documented. This curation process yielded a total of 249 samples, comprising 4,260,545 single cells and 382 distinct proteins. The final dataset thus contained paired RNA and protein expression profiles from each cell for pre-training.

Gene and Protein symbol unification. For the RNA modality, gene symbols in each dataset's raw count matrix were mapped to a reference vocabulary derived from the HUGO Gene Nomenclature Committee (HGNC) database[61]. This vocabulary, comprising 19,264 protein-coding and mitochondrial genes, was selected to match the feature space of the pre-trained scGPT model[12]. To maintain a consistent feature space across all samples, any gene from this reference set not present in a particular sample was included with a count of zero. To resolve the considerable heterogeneity in protein naming conventions across studies, we implemented a comprehensive normalisation pipeline for all proteins (Supplementary Note 2). Common discrepancies included inconsistent prefixes or suffixes derived from fluorophore or batch information (for example, -'CD45RA-BV421' versus 'CD45RA') and variable capitalisation (for example, 'CD3' versus 'Cd3'). To address these inconsistencies, we developed and applied custom scripts using regular expressions to parse and standardise the protein labels. Our pipeline specifically removed substrings following hyphens (e.g., 'CD19-BV786' became 'CD19'), stripped product codes, converted all names to uppercase and manually mapped synonymous markers (e.g., ICOS to CD278). This procedure yielded 382 high-quality, standardised proteins, establishing a robust foundation for subsequent feature alignment and multimodal fusion.

Cell-level quality control and integration. For each dataset, we first ensured that both RNA and protein data were available for the same cell barcodes and retained only these paired cells for subsequent analysis. To filter contaminated empty droplets, extremely low-quality cells and damaged cells, we kept cells with over 200 genes expressed (that is, expression vector with nonzero value count > 200) for pretraining by using the Scanpy[45] packages. Following this quality control step, the processing pipelines for the two modalities diverged. The RNA expression matrix for each cell was normalised to a library size of 10,000 and subsequently log-transformed. The corresponding protein count matrix was separately normalised and then scaled to have zero mean and unit variance, aligning with established common practices for processing protein counts from CITE-seq experiments[33,34].

Finally, the processed data from both modalities were consolidated into a single data object. This object contained key features for each cell, including expression matrices, UMAP coordinates and associated metadata. Where available, cell-type annotations were integrated using a left join on cell identifiers. The final data were stored in the H5MU format, which served as the standard input for model pre-training. The purpose of this pipeline was to generate a standardised and harmonised dataset suitable for the cross-modal pre-training tasks described in the following sections.

CAPTAIN model architecture

We leveraged the dual-encoder structure as the backbone model of CAPTAIN. The transcriptomic module receives four primary inputs: a priori gene knowledge, species tokens, gene tokens, and their corresponding expression values. In parallel, the proteomic module processes cell surface protein tokens. These distinct inputs are transformed by dedicated embedding layers before being encoded by separate transformer architectures. Within the transcriptomic pathway, a masked-attention transformer, inspired by scGPT[12], captures critical gene-gene and gene-cell relationships. The resultant RNA embeddings then guide a cross-attention mechanism within the proteomic encoder to inform the prediction of protein abundances from their corresponding cell surface protein tokens. This RNA-informed process culminates in a unified, multi-omic joint embedding that supports a range of downstream analytical applications (Supplementary Note 4).

Prior biological knowledge. To enrich our pre-training model, we integrated four modalities of prior biological knowledge for 28291 human and 27,443 mouse genes, sourced from the GeneCompass framework, whose embeddings have been demonstrated to enhance the learning of cross-species homology[14]. Each modality was encoded

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into a common 512-dimensional embedding space. The specific embeddings utilised were:

Gene Regulatory Network (GRN) embeddings (emb_{grn}): These were derived from 84 mouse and 76 human GRNs that had been constructed using paired gene expression and chromatin accessibility data from the Encyclopedia of DNA Elements (ENCODE)[62]. The embeddings represent regulatory gene pairs from these networks, generated via the gene2vec algorithm[63].

Promoter embeddings (emb_{pro}): These were generated by fine-tuning a pre-trained DNABert model[64] with promoter sequences spanning 2,500 bp around the transcription start site (500 bp upstream and 2,000 bp downstream).

Gene family embeddings (emb_{fam}): Human gene family data were sourced from the HUGO Gene Nomenclature Committee (HGNC)[61]. Mouse gene families were inferred via homology using BioMart (Ensembl)[65], resulting in 1,645 and 1,539 families for human and mouse, respectively. Within each family, pairs of genes were defined and subsequently embedded using the gene2vec algorithm[63].

Co-expression embeddings (emb_{coe}): These were established from Pearson Correlation Coefficients (PCCs) calculated using uniformly sampled single-cell gene expression data. The final embeddings represent gene pairs with a PCC > 0.8, generated via the gene2vec algorithm[63].

RNA embedding module. Our RNA embedding module adapts the scGPT architecture[12], primarily to leverage its publicly available weights pre-trained on over 33 million cells. This initialisation provides a robust baseline for gene representation. In contrast to the original framework, our module further incorporates specific prior biological knowledge to generate more biologically informed gene embeddings. The module input is derived from the raw cell-by-gene count matrix, denoted as $X \in \mathbb{R}^{N\times G}$. In this matrix, rows correspond to cells ($i = 1, \ldots, N$) and columns to genes ($j = 1, \ldots, G$). For each cell i, its input comprises the following components.

Gene Tokens ($t_g^{(i)}$): In this framework, each gene is treated as a fundamental token of biological information, analogous to a word in natural language processing. A single, unified vocabulary is constructed to enable the harmonisation of data from diverse sources, such as studies employing different sequencing technologies or pre-processing pipelines. This is achieved by taking the union of all unique genes across the relevant datasets, after which each gene g_i in the global vocabulary is assigned a unique integer identifier, $id(g_i)$. A special $<pad>$ token is also introduced to ensure all input sequences have a fixed length through padding. Following this tokenization, each cell i is represented as a sequence of M integer identifiers in a vector $t_g^{(i)} \in \mathbb{N}^M$:

$$t_g^{(i)} = \left[id\left(g_1^{(i)}\right), id\left(g_2^{(i)}\right), \ldots, id\left(g_M^{(i)}\right)\right] \tag{1}$$

where M is a predefined hyperparameter for the maximum input length.

Species Tokens ($t_s^{(i)}$): To explicitly inform the model of the species of origin for each cell, a species-specific token is incorporated into the input representation. This enables the model to learn and leverage species-specific biological patterns during pre-training. The species identity is encoded using a simple integer mapping, where human is assigned an identifier of 0 and mouse an identifier of 1. This identifier is then broadcast across the maximum sequence length, M, to create a species token vector, $t_s^{(i)} \in \mathbb{Z}_2^M$, for each cell i. This vector is defined as $t_s^{(i)} = [s_i, s_i, \ldots, s_i]$, where the species identifier s_i is determined by the origin of cell i:

$$s_i = \begin{cases} 0, \text{ if cell } i \text{ is from Human,} \\ 1, \text{ if cell } i \text{ is from Mouse.} \end{cases} \tag{2}$$

Gene expression values (x^i): To address the variability in expression magnitudes arising from factors like sequencing depth and gene sparsity, the value binning technique is adopted[12]. This method normalises expression counts onto a relative, cell-specific scale. The procedure is applied on a per-cell basis. To speed up the pre-training, we restrict the input to only genes with non-zero expression for each input cell. For each cell i, its non-zero expression values $X_{i,j}$ ($X_{i,j} > 0$) are sorted and partitioned into B quantiles of equal size. This process defines B consecutive, cell-specific intervals $[b_k, b_{k+1}]$, where $k \in 1, 2, \ldots, B$. The raw expression count $X_{i,j}$ is then mapped to a binned integer value $x_j^{(i)}$ according to the rule:

$$x_j^{(i)} = \begin{cases} k, \text{if} X_{i,j} > 0 \text{and} X_{i,j} \in [b_k, b_{k+1}], \\ 0, \text{if} X_{i,j} = 0. \end{cases} \tag{3}$$

This ensures the resulting values have a consistent semantic meaning; for instance, a binned value of B consistently represents the highest expression tier within that specific cell. For certain fine-tuning applications, this binning step is preceded by log1p transformation and highly variable gene (HVG) selection. The final vector of binned values for cell i is denoted as $x^{(i)} = [x_1^{(i)}, x_2^{(i)}, \ldots, x_M^{(i)}]$.

Input Embeddings: The gene tokens and species tokens, emb_g and emb_s, are embedded using conventional pytorch embedding layers (torch.nn.Embedding) to facilitate the mapping of each token to a fixed-length embedding vector of dimension D. Concurrently, binned expression values are processed by a multi-layer perceptron (emb_b), which captures the ordinal nature of gene expression values. The final embedding for cell i, $h^{(i)} \in \mathbb{R}^{M\times D}$, is the element-wise sum of these component vectors:

$$h^{(i)} = emb_g\left(t_g^{(i)}\right) + emb_b\left(x^{(i)}\right) + emb_s\left(t_s^{(i)}\right) + emb_{grn} + emb_{pro} + emb_{fam} + emb_{coe} \tag{4}$$

Furthermore, while our prior knowledge is comprehensive, encompassing nearly all known human and mouse genes, any gene not present in this pre-defined vocabulary is assigned a zero vector for its embedding.

Protein embedding module. This module encodes a panel of 382 curated cell surface protein tokens. Each token is assigned a unique integer identifier, $id(p_j)$, to form a vocabulary. A special classification token, $<cls>$, is included to aggregate the protein-level embeddings into a global cell representation. To ensure a standardized input format, every cell is represented using this complete, fixed vocabulary. The expression for any protein not measured in a cell is set to 0, indicating missing measurement rather than biological absence. For a given cell, i, the sequence of its protein tokens is represented as a vector, $t_p^{(i)} \in \mathbb{N}^P$:

$$t_p^{(i)} = \left[id\left(p_1^{(i)}\right), id\left(p_2^{(i)}\right), \ldots, id\left(p_P^{(i)}\right)\right] \tag{5}$$

where P=382. These protein tokens are embedded using conventional PyTorch embedding layers, $q^{(i)} = emb_p(t_p^{(i)})$, which project each discrete identifier into a fixed-length embedding vector of dimension D.

RNA encoder module. To address the non-sequential nature of single-cell data and effectively leverage the pre-trained parameters derived from scGPT[12], we employed a masked-attention transformer architecture to encode the complete input RNA embeddings, denoted as $h^{(i)}$ (Equation (4)). The transformer's self-attention mechanism operates on sequences of M embedding vectors, allowing it to capture putative gene-gene interactions. The output of the final n_{th} transformer layer, the resulting RNA representation, $h_n^{(i)} = Transformer(h^{(i)}) \in \mathbb{R}^{M\times D}$, is then fused with corresponding protein embeddings via a cross-attention module to generate a robust multi-modal embedding. To ensure computational efficiency when processing large gene panels (where the input dimension M can be in the tens of thousands), the model employs an accelerated self-attention mechanism, FlashAttention. This memory-efficient implementation allows for the feasible and effective application of self-attention to extensive genomic data.

Furthermore, standard autoregressive models are ill-suited for gene expression data, which, unlike natural language, is non-sequential and lacks an inherent order for prediction. To address this challenge, we utilized the specialized attention masking strategy to enable autoregressive-style generation. The approach is to predict expression values for a subset of 'unknown' genes based on a set of 'known' genes. This is governed by a custom attention mask, A_{mask}, which modifies the standard attention equation:

$$Q = h_{n-1}^{(i)} W_q, K = h_{n-1}^{(i)} W_k, V = h_{n-1}^{(i)} W_v \tag{6}$$

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d}} + A_{mask}\right)V \tag{7}$$

where $Q, K, V \in \mathbb{R}^{M \times d}$ represent the query, key and value vectors. $W_q, W_k, W_v \in \mathbb{R}^{D \times d}$ are the learnable weight matrices. Specifically, the mask prohibits a given query from attending to any 'unknown' genes other than itself by setting the corresponding weights in the attention matrix to $-\infty$. This is defined for each element $a_{i,j}$ in A_{mask} as:

$$a_{ij} = \begin{cases} -\infty, & \text{if} i \neq j \text{and} j \in \text{unknowngenes} \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

Generation is performed iteratively. In each step, the model predicts expression values, and the most confidently predicted genes are added to the 'known' set for the next iteration. This workflow effectively mimics causal prediction for non-sequential data and unifies generation from both gene-level and cell-level prompts.

Protein encoder module. The protein encoder module uses a canonical Transformer-encoder architecture, augmented by a dedicated cross-attention block to fuse RNA and protein information into a unified representation. The parameters of this module are initialised at random. Given an initial protein embedding matrix, $q_0^i \in \mathbb{R}^{P \times D}$, where P is the number of protein tokens, and D is the embedding dimension, we first apply a standard self-attention and feed-forward layer. For n_{th} layer, the output of the self-attention mechanism, $SelfAttention_n$, is computed from the previous layer's output, $q_{n-1}^{(i)}$:

$$Q_{self} = q_{n-1}^{(i)} W_q, K_{self} = q_{n-1}^{(i)} W_k, V_{self} = q_{n-1}^{(i)} W_v \tag{9}$$

$$SelfAttention(Q_{self}, K_{self}, V_{self}) = softmax\left(\frac{Q_{self} K_{self}^T}{\sqrt{d}}\right) V_{self} \tag{10}$$

The output of the attention head is then passed through a residual connection, layer normalization, and a standard position-wise feed-forward network. After such block, the final output is $q_n^{(i)}$, which represents an embedding that encodes intra-protein relationships.

After that, to integrate RNA context, we introduce a cross-attention layer that uses the precomputed RNA embedding $h^{(i)} \in \mathbb{R}^{M \times D}$ as both queries and keys, while treating the protein-encoded features $q^{(i)} \in \mathbb{R}^{P \times D}$ as values. Concretely, we compute:

$$Q_{cross} = h_n^{(i)} W_q, K_{cross} = h_n^{(i)} W_k, V_{cross} = q_{n-1}^{(i)} W_v \tag{11}$$

$$CrossAttention(Q_{cross}, K_{cross}, V_{cross}) = softmax\left(\frac{Q_{cross} K_{cross}^T}{\sqrt{d}}\right) V_{cross} \tag{12}$$

This cross-attention block produces a matrix of size $\mathbb{R}^{P \times d}$, which is then projected back to $\mathbb{R}^{P \times D}$, merged with $q_{n-1}^{(i)}$ via a residual connection, and normalized. The result is a joint RNA-protein embedding $q^{(i)} \in \mathbb{R}^{P \times D}$ that simultaneously captures transcriptomic and proteomic features, ready for downstream tasks.

Cell-level representation. A unified, cell-level representation, $h_c^{(i)}$, is generated by learning to aggregate features across both the transcriptomic and proteomic modalities. To facilitate this learned aggregation, we introduce a special classification token, $<cls>$, which is prepended to the input sequence of protein tokens. This token, along with the protein features, passes through the protein-specific self-attention encoder and is subsequently fused with the RNA embedding via the cross-attention module. The final hidden state from the fusion block corresponding to this $<cls>$ token is then taken as the definitive multi-modal cell representation. This single vector, $h_c^{(i)} \in \mathbb{R}^D$, thereby encapsulates the salient, cross-modally informed features of the cell for all downstream predictive tasks.

Learning objective for pretraining

The model is pretrained using a composite objective function designed to learn multi-modal representations from single-cell data. This objective combines three distinct components: (1) a masked gene expression prediction task within the RNA modality; (2) a cross-modal prediction of protein expression from RNA features; and (3) uncertainty quantification for the protein predictions via interval estimation. Collectively, these components enable the model to learn both intra-modal gene regulatory patterns and inter-modal transcript-protein relationships.

Masked gene expression prediction: The masked gene expression (MGE) prediction objective is utilized to learn gene-gene interdependencies. This task requires the model to infer the expression levels of a randomly masked subset of genes based on the expression of the remaining, unmasked genes within a given cell. During training, the expression values for a random fraction of genes in each cell, i, are masked. The model's RNA encoder then processes the available transcriptomic information to produce contextualised embeddings. Subsequently, a multi-layer perceptron (MLP) head is used to predict the expression value, $x_j^{(i)}$, for each masked gene, j, from its corresponding hidden state representation, $h_j^{(i)}$. The MGE objective is to minimise the mean squared error (MSE) between the predicted and actual expression values. This loss function, denoted L_{MGE}, is defined as:

$$L_{MGE} = \frac{1}{\left|U_{unk}^{(i)}\right|} \sum_{j \in U_{unk}^{(i)}} \left(MLP\left(h_j^{(i)}\right) - x_j^{(i)}\right)^2 \tag{13}$$

Here, U_{unk} is the set of indices for the masked genes in cell i; $x_j^{(i)}$ is the ground-truth expression for the $j-th$ masked gene; and $MLP(h_j^{(i)})$ is the value predicted by the MLP. The denominator $|U_{unk}^{(i)}|$ is the total number of masked genes. Optimising this objective enables the model to learn the biological relationships and dependencies within the gene expression data.

Protein expression prediction: To align transcriptomic and proteomic representations, the model is trained to predict protein abundance from the joint embedding space. Specifically, for each cell i, a final joint embedding, z^i, is produced by fusing the representations from both the RNA and protein encoders. An MLP prediction head then takes this joint embedding z^i as input to predict the expression level for every protein, j, that was measured in that cell's panel (denoted $P^{(i)}$. The predicted expression for protein j is denoted as

$MLP(z^{(i)})_j$. The learning objective is to minimise the MSE between these predictions and the ground-truth protein expression levels, $y_j^{(i)}$, This cross-modal prediction loss, L_{PE}, is therefore defined as:

$$L_{PE} = \frac{1}{\left|P^{(i)}\right|} \sum_{j \in P^{(i)}} \left(MLP(z^{(i)})_j - y_j^{(i)} \right)^2 \tag{14}$$

Here, $\left|P^{(i)}\right|$ is the number of proteins measured in cell i. By calculating the loss exclusively over the set of available proteins, this objective robustly handles the incomplete protein panels common in CITE-seq data. Optimising this objective compels the model to learn a joint representation where transcriptomic states are predictive of proteomic outcomes.

Protein expression interval prediction: We are interested not only in predicting protein expression but also in quantifying the uncertainty of our prediction using interval estimation. To that end, we will want to estimate not only the mean expected protein expression given the RNA expression profile of the cell but also a vector of quantiles that can be used to construct prediction intervals. Let $Q = \{q_1, q_2, \ldots, q_k\}$ be the set of quantiles we wish to estimate. Then we want to estimate the protein expression y and the predicted protein expression quantile $\hat{y}_q$ for $q \in Q$ such that we minimize the following objective:

$$L_q\left(y, \hat{y}_q\right) = \left(I\left(\hat{y}_q > y\right) \times (1 - q) + I\left(\hat{y}_q < y\right) \times q \right) \left|\hat{y}_q - y\right| \tag{15}$$

$$L_{\mathrm{PEI}}(y, \hat{\sigma}) = \frac{1}{k} \sum_{q \in Q} L_q\left(y, \hat{y}_q\right) \tag{16}$$

where I represents the indicator function. The role of the indicator function here is to apply an asymmetric penalty. It selectively uses either the weight q or $1 - q$ depending on whether the prediction is an overestimation or an underestimation, which is the core principle of quantile regression.

Fine-tuning on downstream tasks

For the single-cell protein imputation and expansion tasks, datasets from various tissues were randomly partitioned into training (90%) and test (10%) subsets at the cell level. We retained the intersection of gene markers common to both our pre-trained foundation model and the reference set. Prior to fine-tuning, all data preprocessing was performed using the Scanpy package. Gene expression values were normalised, log-transformed, and subsequently binned. For the paired protein modality, to mitigate batch effects across diverse tissues, samples, and disease states, we applied a specific cell-level normalization strategy following Lakkis et al.[34]. Specifically, the expression of a given protein in each cell was divided by the total protein expression in that cell, multiplied by the median total expression for that protein across the specific dataset, and then transformed to a natural log scale. Consequently, CAPTAIN takes these preprocessed values as input and predicts protein expression in the same log-normalized space. Only gene markers with non-zero expression values were utilised for training.

In the fine-tuned setting (protein imputation), CAPTAIN was fine-tuned on a subset of paired RNA-protein cells from the target dataset and then used to infer proteins within the fixed 382-protein vocabulary for held-out cells, including proteins not measured in a given antibody panel. In the zero-shot setting (protein expansion), CAPTAIN directly predicted proteins within the same fixed vocabulary from transcriptomic inputs without any dataset-specific fine-tuning. Here, the model predicted the expression of all proteins in the predefined standardized surface protein vocabulary using only the cellular transcriptome as input, without requiring paired protein measurements for the test cells. For both applications, the model was initialised with our pre-trained weights. The fine-tuning process mirrored the parameter settings of the pre-training phase: we used a learning rate of 1e-4 with a decay to 90% of its value after each epoch, and a batch size of 20.

We benchmarked CAPTAIN against five contemporary methods for single-cell protein prediction: sciPENN[34], Seurat (v4)[33], TotalVI[36], scTranslator[37] and scTEL[38]. For Seurat and TotalVI, we followed their standard, published prediction workflows for the comparison. To evaluate performance, we calculated two key metrics for each protein by comparing its measured and predicted expression values across all cells: the Pearson correlation coefficient, to assess the linear relationship, and the Root Mean Square Error (RMSE), to quantify the magnitude of prediction error. For the RMSE, to ensure a fair comparison across methods with varying output scales (e.g., raw counts vs. denoised values), we z-score standardized each protein across test cells using statistics computed on the test set before computing RMSE.

Cell type annotation. For the cell type annotation task, datasets from diverse tissues with manually curated cell type labels were randomly partitioned into training (90%) and test (10%) subsets. This procedure was repeated using multiple random seeds to ensure reproducibility. The model was subsequently fine-tuned on the training data, and its annotation performance was evaluated on the held-out test data. The input features were restricted to the intersection of gene markers common to our pre-trained foundation model and the reference dataset. Prior to fine-tuning, we used Scanpy to normalise, log-transform, and bin gene expression values, and to separately normalise and scale paired protein expression values. All gene markers, including those with zero expression, were retained for the training process.

Our model is designed to perform cell type annotation on two distinct data modalities: single-cell transcriptomes alone (scRNA-seq) and paired single-cell transcriptomic and proteomic data (e.g., CITE-seq). The core architecture remains consistent, with the primary distinction being the placement of the output classifier. For scRNA-seq data, a multilayer perceptron (MLP) classifier is appended to the final layer of the RNA encoder module. For paired multi-omic data, this MLP classifier is instead connected to the final layer of the protein encoder module. To begin the fine-tuning process, the model was initialised with weights from the pre-trained foundation model. The only exception was the output MLP classifier, whose weights were randomly initialised. Fine-tuning was initiated with a learning rate of 1e-3 and a batch size of 26. The learning rate was scheduled to decay after each epoch to 90% of its value from the previous epoch. The training objective was to minimise the cross-entropy loss between the predicted cell type probabilities, derived from the learned cellular representations, and the ground-truth cell type labels.

We benchmarked the performance of our model against two contemporary cell type annotation methods: scGPT[12] and Seurat (v4)[33]. The comparison was conducted on multiple datasets using a five-fold cross-validation scheme. Performance was assessed using four standard classification metrics: *Accuracy*, *Precision*, *Recall*, and the *Macro-F1* score. *Accuracy*, *Precision*, and *Recall* were calculated globally to provide a measure of overall performance. In contrast, the *Macro-F1* score was calculated as the unweighted mean of the *F1* scores for each cell type class, thereby giving greater importance to the correct classification of rare cell populations (Supplementary Note 3). For the scGPT comparison, we used the standard scGPT model for scRNA-seq-only datasets and the specific scGPT variant designed for multi-omic integration when analysing CITE-seq data. Both models were trained and used for prediction with default parameters. For the Seurat comparison, we employed distinct workflows depending on the data modality. For scRNA-seq data, we utilised the TransferData function. This method identifies anchors between the reference (training) and query (test) datasets based on principal component analysis (PCA) of the training data, subsequently transferring the known cell type labels to the test set. For multi-omic datasets, we first applied the weighted

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nearest neighbour (WNN) analysis to learn a joint representation of the RNA and protein modalities. Following this integration, we applied the TransferData workflow on the resulting integrated space to transfer labels from the training to the test set. In both cases, the transferred labels were taken as the predicted cell types for the test data.

Multi-omics integration. For the multi-omic integration task, our model focused on integrating single-cell transcriptomics and proteomics. To evaluate its performance robustly, we used both single-batch and multi-batch multi-omic datasets. Each dataset was also partitioned using a 9:1 ratio. The preprocessing pipelines for the single-cell transcriptomic and proteomic data were identical to those described in the preceding tasks. For the RNA encoder module, the 3,000 most highly variable genes (HVGs) with non-zero expression were selected for training.

We initialised the model with all parameters loaded from the pre-trained foundation model. The final multi-omic representation for each cell was derived from the embedding of the special < *cls* > token within the protein encoder module. Notably, to facilitate both batch effect removal and effective integration, we adopted several fine-tuning objectives for the RNA encoder module as proposed by scGPT. These included Gene Expression Prediction (GEP), Gene Expression Prediction for Cell modelling (GEPC), and Domain Adaptation via Reverse back-propagation (DAR)[12]. The fine-tuning process was initiated with a learning rate of 1e-3, which was scheduled to decay to 95% of its value after each epoch. The weight for the DAR objective was set to the default value of 1.0. The model was trained for a fixed 25 epochs with a batch size of 16.

We benchmarked CAPTAIN against recent multi-omic integration methods, including scGPT[12], Scanpy (MOFA)[66], and TotalVI[36], on both single-batch and multi-batch datasets. For a fair comparison, all four methods were trained using an identical input feature set, comprising 3,000 HVGs from the transcriptomic data and all available proteins from the proteomic data. The three competing methods were run using their publicly available, default workflows.

To evaluate data integration performance, we adopted the comprehensive suite of metrics established in the benchmarking study by Luecken et al.[67] and scGPT[12]. These metrics assess two key aspects of integration quality: the preservation of biological heterogeneity and the degree of batch effect removal (Supplementary Note 3). First, to evaluate the preservation of biological identity, we used four biological conservation metrics. Normalized Mutual Information (NMI_{cell}) quantifies the concurrence between the ground-truth cell type labels and the Louvain cluster labels derived from the integrated embeddings. Louvain clustering was performed across resolutions from 0.1 to 2 (in increments of 0.1), and the best score was selected. A higher NMI_{cell} score, ranging from 0 to 1, indicates a better match. The Adjusted Rand Index (ARI_{cell}) was also used to measure the agreement between annotated labels and the NMI-optimized Louvain clusters, adjusted to account for chance. The ARI_{cell} score ranges from 0 (random labelling) to 1 (perfect match). The Average Silhouette Width (ASW_{cell}) assesses cluster separation by comparing a cell's within-cluster distance to its distance from the nearest neighbouring cluster. We scale the raw ASW score for cell types (C) to a 0–1 range using the formula:

$$ASW_{cell} = \left(ASW_C + 1\right)/2 \tag{17}$$

where 1 indicates well-separated clusters. Finally, we computed $AvgBIO$, the aggregate average of these three metrics:

$$AvgBIO = \left(ARI_{cell} + NMI_{cell} + ASW_{cell}\right)/3 \tag{18}$$

Second, to evaluate integration performance on multi-batch datasets, we used two batch-correction metrics. The ASW_{batch} metric is calculated based on batch labels (B) and adjusted to reflect mixing, where a higher score indicates better batch integration:

$$ASW_{\text{batch}} = 1 - \left|ASW_B\right| \tag{19}$$

The Graph Connectivity ($GraphConn$) metric quantifies the connectedness of cells of the same type within a k-nearest neighbours (kNN) graph. It is calculated as the average fraction of cells belonging to the largest connected component (LCC) for each cell type ($c \in C$):

$$GraphConn = \frac{1}{|C|}\sum_{c\in C}\frac{\left|LCC\left(G_c^{kNN}\right)\right|}{N_c} \tag{20}$$

To summarize batch-mixing performance, we calculated $AvgBATCH$, representing the average of ASW_{batch} and $GraphConn$:

$$AvgBATCH = \left(ASW_{\text{batch}} + \text{GraphConn}\right)/2 \tag{21}$$

Finally, we also report an *Overall* score, calculated as the weighted average of the biological conservation and batch correction scores: $0.6 \times AvgBIO + 0.4 \times AvgBATCH$.

$$Overall = 0.6AvgBIO + 0.4AvgBATCH \tag{22}$$

Batch correction. When combining datasets from different sequencing batches or technologies, batch effects can be a major issue for downstream analysis, such as cell type clustering. Our goal was therefore to correct for this technical noise in both single-modality and multi-modality datasets, while preserving the true biological variance. All datasets were split into training (90%) and test (10%) subsets. The preprocessing method for both single-cell transcriptomic and proteomic data was the same as in the previously described tasks. For model input, we also used 3000 HVGs with non-zero expression for the RNA encoder module and all available proteins for the protein encoder module.

To correct for batch effects, we fine-tuned the model after loading all pre-trained weights. The cell representation was taken from different sources depending on the input data. For single-modality transcriptomic data, the embedding of the < *cls* > token from the RNA encoder module was used. For multi-omic datasets, this representation was instead taken from the < *cls* > token of the Protein encoder module. We used a multi-task fine-tuning strategy to ensure effective batch integration. We continued to use the GEP and GEPC objectives in the multi-omic integration task. In addition, we added three more objectives to improve performance: Elastic Cell Similarity (ECS), as well as DAR and Domain-Specific Batch Normalization (DSBN) for direct batch correction[12]. Following the default parameters for these objectives, the masking ratio for GEP and GEPC was set to 0.4, and the β parameter in ECS was set to 0.6. Fine-tuning was started with a learning rate of 1e-3, and the learning rate decayed to 90% of its value after each epoch. The model was trained for a fixed 15 epochs using a batch size of 16. We compared the batch correction performance of CAPTAIN against three other common methods: scGPT[12], Seurat[68], and Harmony[40]. To ensure a thorough evaluation, performance was measured using the same set of biological conservation and batch-correction metrics as detailed in the multi-omic integration task.

Cell-cell communications inference. Our cell-cell communication inference method enhances biological fidelity by utilising distinct molecular modalities: transcriptomic data for ligand expression and proteomic data for receptors. This strategy directly addresses the often poor correlation between mRNA and protein levels and leverages direct cell surface protein quantification (or robust model-based predictions thereof) for a more accurate assessment of cellular receptive capacity. Specifically, ligand expression is derived from scRNA-seq. The requisite cell surface receptor proteomic data are obtained either via zero-shot prediction from scRNA-seq alone or through model fine-tuning and imputation when partial proteomic measurements (e.g., from CITE-seq) are available. Communication pathways are subsequently inferred using these modality-specific profiles and a curated consensus ligand-receptor interaction database (e.g., incorporating resources such as CellPhoneDB[69], CellChat[52], ICELLNET[70], connectome[51], and CellTalkDB[71]). A ligand (from RNA data) or a receptor (from protein data) is considered expressed within a cell type if its molecular abundance meets defined criteria (e.g., a 30% threshold percentage of expressing cells). For multimeric receptors, the co-expression of all constituent subunits, assessed at the protein level, is required for the receptor to be considered functionally present. The potential strength of interactions between cell populations is quantified based on the mean expression of participating ligands (RNA-derived) and their cognate receptors (protein-derived), accounting for the limiting subunit (protein level) in multimeric receptor complexes. Statistical significance is determined via a permutation test (e.g., 1,000 permutations), wherein cell population labels are randomly shuffled to generate an empirical null distribution for each ligand-receptor interaction score. Interactions yielding an empirical p-value below a defined significance threshold (e.g., $p < 0.05$) are deemed statistically significant. This process yields a network of putative intercellular signalling pathways robustly grounded in our modality-specific molecular assessment.

To infer intercellular signalling pathways, we first establish comprehensive single-cell transcriptomic and proteomic profiles. When only scRNA-seq data are available, our model performs zero-shot prediction of protein expression, leveraging its pre-established understanding of transcript-protein relationships. Alternatively, where paired scRNA-seq and partial proteomic data are provided, the model is fine-tuned to enhance predictions and impute expression for proteins not directly measured, thereby enriching the proteomic information for subsequent communication analysis. This strategy directly addresses the often-poor correlation between mRNA and protein levels and leverages direct cell surface protein quantification (or robust model-based predictions thereof) for a more accurate assessment of cellular receptive capacity.

Cell-cell communication is then inferred from these integrated profiles using a curated ligand-receptor interaction database, which is a consensus resource composed by multiple expert-curated ligand-receptor resources, including CellPhoneDB[69], CellChat[52], ICELLNET[70], connectome[51], and CellTalkDB[71]. A ligand or receptor is considered expressed within a cell type if its expression meets defined criteria (e.g., a minimum expression level and percentage of expressing cells). For multimeric receptors, the co-expression of all constituent subunits within the same cell population is required.

The potential strength of interactions between cell populations is quantified based on the mean expression of the participating ligands and their cognate receptors, accounting for the limiting subunit in multimeric receptor complexes. Statistical significance is assessed via a permutation test (e.g., 1000 permutations), wherein cell population labels are randomly shuffled to generate an empirical null distribution for each ligand-receptor interaction score. An empirical p value is subsequently derived by comparing the observed interaction score against this null distribution. Interactions falling below a defined significance threshold (e.g., $p < 0.05$) are deemed statistically significant, collectively forming a network of putative intercellular signalling pathways.

Implementation details

All experiments were conducted on institutional high-performance computing resources. The foundation model for transcriptomic data was configured with an embedding dimension of 512. The architecture consists of 12 stacked Transformer blocks, each with 8 attention heads, and a hidden layer size of 512 in the fully-connected networks. For the proteomics-based foundation model, the embedding dimension was also 512, but the architecture comprised 6 stacked Transformer blocks, each with 8 attention heads, and a consistent hidden layer size of 512.

During pre-training of the transcriptome module, we set a maximum input sequence length of 1200 genes. For cells with more than 1200 non-zero expressed genes, a random subset of 1200 genes was sampled as input at each training iteration. A masking ratio of 0.15 was applied to the transcriptomic data. The proteomic module used a fixed input length of 382, corresponding to the total number of protein tokens.

The models were optimized using the Adam optimizer with a batch size of 20 and an initial learning rate of 1e-4. The learning rate was multiplied by a factor of 0.9 after each epoch. Pre-training was conducted for a total of 10 epochs. The entire pre-training process on over 4.2 million cell pairs ran continuously for approximately 7 days on a high-performance computing system equipped with four NVIDIA H800 GPUs (80GB VRAM each), 5 TB of RAM, and 437 TB of disk storage. During this phase, the peak GPU memory usage was approximately 75 GB per accelerator, and the peak system RAM usage was 100 GB. Importantly, this extensive computational cost applies exclusively to the one-time pre-training phase. For practical downstream applications, a direct comparison against baseline foundation models demonstrates that CAPTAIN is highly efficient; its modest computational overhead for dual-omics processing is highly competitive and well-justified by the substantial performance advantages (Supplementary Data 21).

To prevent data leakage, the fine-tuning and test datasets for the protein inference task were constructed to be completely disjoint from the pre-training dataset. For other tasks, namely cell type annotation, batch integration, multi-omics integration, and cell-cell communication inference, the fine-tuning and test sets partially overlapped with the pre-training data. This design choice was explicitly justified by the scarcity of paired multi-omics datasets with task-specific labels, which were not exposed during the pre-training phase.

For all downstream tasks, we utilized the same model layer configurations as in pre-training, with a batch size of 15. Each dataset was partitioned into training and validation sets at a 9:1 ratio. The learning rate for the protein inference task was initiated at 1e-4, while for all other tasks, it was set to 1e-3. In all cases, the learning rate was decayed by 10% after each epoch. Models were trained for a fixed 15 epochs, and the model checkpoint yielding the best performance on the validation set was saved for final evaluation on the test set. For integration tasks, parameters within the transcriptome module followed the scGPT defaults: the masking ratio for GEP and GEPC was set to 0.4. The β parameter in ECS was set to 0.6, and the ECS loss was assigned a weight of 10 when combined with other loss functions. A DAR weight of 1.0 was used for batch integration.

The CAPTAIN neural network model was implemented using PyTorch. Preprocessing of transcriptomic and proteomic expression data, including normalization, log-transformation, and highly variable gene selection, was performed using the Muon Python library. Evaluation metrics for batch integration and multi-omics integration tasks were calculated using implementations from scib.metrics, while metrics for the cell annotation task were implemented with scikit-learn. Comparative analysis for cell-cell communication inference was conducted using LIANA+, and pathway enrichment analysis was performed with the GSEApy package. Key software dependencies include torchtext 0.16.2, torch-geometric 2.6.1, flash-attn 1.0.1, pandas 2.0.3, omicverse 1.6.10, umap-learn 0.5.7, and h5py 3.12.1.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The curated pre-training datasets (scT & P-4M) generated in this study are available at our dedicated data portal [https://sctp4m.aigenomicsyulab.com]. A comprehensive overview of all datasets used for pre-training, including modality, species, cell/protein counts, tissue type, disease context, and data source, is provided in Supplementary Data 1. For the cell surface protein imputation and expansion task, the human PBMC data used in this study are available in the GEO database under accession code https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE164378GSE164378, and the mouse PBMC data are available in the GEO database under accession code https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE193733GSE193733. The human MALT data are available at https://www.10xgenomics.com/resources/datasets/10-k-cells-from-a-malt-tumor-gene-expression-and-cell-surface-protein-3-standard-3-0-0. The human MNC data are available at https://upenn.box.com/s/64c9fsex50g1bhv67893cpdg9c5jqjzo. For the cell type annotation task, the multi-modal human PBMC data used in this study are available in the GEO database under accession code https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE164378GSE164378, the multi-modal COVID-19 data under accession code https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE161918GSE161918, and the BMMC dataset (combining CD4 + and CD8^{+} T cells) under accession code https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE194122GSE194122. Additional datasets for this task are available at the following URLs: the 10x Multiome PBMC data at https://support.10xgenomics.com/single-cell-multiome-atac-gex/datasets/1.0.0/pbmc_granulocyte_sorted_10k, the Multi-tissue Immune Cells data at https://www.tissueimmunecellatlas.org, and the Myasthenia Gravis dataset at https://singlecell.broadinstitute.org/single_cell/study/SCP1532. For the multi-batch integration task, the datasets used in this study are available in the GEO database under accession codes https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE164378GSE164378 (Healthy PBMCs), https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE161918GSE161918 (SARS-CoV-2 PBMCs), and https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE194122GSE194122 (Healthy BMMCs). For the multi-omic integration task, the bone marrow mononuclear cell data used in this study are available in the GEO database under accession code https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE194122GSE194122, and the composite single-cell datasets are available under accession codes https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE158013GSE158013 (TEA-seq data) and https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE164378GSE164378 (ECCITE-seq and CITE-seq data). For the cell-cell communication inference task, the PBMC CITE-seq data used in this study are available in the GEO database under accession code https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE164378GSE164378, and the independent PBMC data under accession code https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE161918GSE161918. The multi-sample COVID-19 data are available at https://covid19cellatlas.org/. Source data are provided with this paper. All data supporting the findings of this study are available within the paper and its Supplementary Information. Source data are provided with this paper.

Code availability

The code[72] used in the study is publicly available from the GitHub repository: https://github.com/iamjiboya/CAPTAIN/tree/main.

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Author contributions

F.Y. and B.J. conceived the project and designed the algorithm. B.J. implemented the algorithm. T.H. and J.W. constructed the scT & P-4M dataset. B.J., T.H., J.W., M.L., L.X., Q.Z., L.Q., Y.Z., and S.P. performed computational experiments and contributed to data interpretation. S.Z. developed the companion website. B.J., T.H., J.W., and F.Y. wrote the manuscript with input from all authors. F.Y., Y.Z., and S.P. supervised and directed the study.

Competing interests

The authors declare no competing interests.

Additional information

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